

# A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot

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**Abstract**—With the increasing demand for autonomous underwater operations in complex environments, autonomous obstacle avoidance has become a fundamental capability for biomimetic eel-like robots. However, such platforms exhibit strongly coupled locomotion characteristics, where propulsion and steering are difficult to directly decouple, while autonomous control for this type of biomimetic platform remains relatively underexplored. To address these challenges, this paper proposes a two-stage learning framework for autonomous obstacle avoidance of an eel-like robot, where imitation learning is first used for policy initialization and reinforcement learning is then adopted for policy fine-tuning. To better match the locomotion characteristics of the platform, AHTA-CPG is incorporated into the autonomous control framework to achieve decoupled control of steering and propulsion. A MuJoCo-based simulation environment is constructed from the physical prototype, and domain randomization is adopted to reduce the domain gap between simulation and real underwater images. Simulation and real-world experiments demonstrate that the proposed method achieves stable obstacle-avoidance performance and shows good practical deployment capability.

**Index Terms**—Biomimetic eel-like robot; Autonomous obstacle avoidance; Imitation learning; Reinforcement learning; AHTA-CPG

## I. INTRODUCTION

With the increasing demand for autonomous underwater operations in complex environments [1], autonomous obstacle avoidance has become essential for safe operation and sustained task execution. Compared with conventional propeller-driven platforms, biomimetic eel-like robots use continuous body-wave propulsion and compliant structures, giving them strong maneuvering potential in narrow, confined, and obstacle-dense environments. However, their locomotion is highly coupled, and propulsion and steering are less directly controllable than those of propeller-driven robots, which makes autonomous control challenging [2], [3].

Existing studies on biomimetic eel-like robots have mainly focused on dynamic modeling, undulatory propulsion, and low-level swimming control. These works have laid an important foundation for understanding eel-like robot motion and developing biomimetic propulsion mechanisms, including dynamic modeling studies [4], [5] and rhythm-based control

methods using central pattern generators (CPGs) [6]. Yu et al. further proposed AHTA-CPG, which modulates tail oscillation rhythms through head deflection and provides a control method suited to the platform characteristics of eel-like robots [7]. Overall, these studies mainly address biomimetic swimming, while autonomous control in complex obstacle-rich environments remains less explored.

Different from low-level swimming control, autonomous obstacle avoidance is a perception-driven continuous decision-making problem. The robot must infer obstacle distribution and feasible motion directions from environmental observations, while generating behaviors that balance safety and locomotion performance. Deep reinforcement learning can directly learn perception-to-action mappings through interaction, offering a feasible solution to this problem [8]. For biomimetic eel-like robots, however, two challenges remain. First, policy learning is difficult because of unstable training and slow policy formation. Second, the learned policy must operate on a strongly coupled biomimetic locomotion mechanism, which further increases the difficulty of control adaptation. Therefore, developing an effective learning-based method for autonomous obstacle avoidance on such platforms is the core problem of this paper.

To address this problem, this paper proposes a two-stage learning framework for autonomous obstacle avoidance of a biomimetic eel-like robot. The method first initializes the policy through imitation learning, enabling the robot to acquire basic obstacle-avoidance behaviors, and then refines the policy through reinforcement learning to improve adaptability and control performance in complex environments. In control implementation, AHTA-CPG is incorporated as the low-level controller, allowing policy learning to build on a control process consistent with the platform characteristics. A MuJoCo-based simulation environment is constructed for training and validation, and domain randomization is adopted to facilitate real-world deployment. Simulation and real-world experiments show that the proposed method achieves stable and effective autonomous obstacle avoidance with practical deployment capability.

The main contributions of this work are as follows:

- 1) A two-stage learning framework is proposed for autonomous obstacle avoidance of a biomimetic eel-like robot. The framework adopts a training pipeline of imitation learning initialization followed by reinforcement

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learning fine-tuning, thereby improving the feasibility and stability of policy learning.

- 2) AHTA-CPG is incorporated into the autonomous control framework as the low-level control method, so that the learned policy can better adapt to the biomimetic propulsion process of the eel-like robot.
- 3) Simulation and real-world experiments are conducted to validate the effectiveness of the proposed method in autonomous obstacle avoidance, demonstrating its stable avoidance performance and practical deployment capability.

## II. METHOD

This section presents the proposed two-stage learning framework for autonomous obstacle avoidance of the biomimetic eel-like robot. It first introduces the robotic platform and its locomotion characteristics, followed by a MuJoCo-based simulation environment constructed from the physical prototype for policy training. On this basis, the two-stage learning method is described, including imitation learning initialization and reinforcement learning fine-tuning. Finally, domain randomization is adopted to improve the deployability of the learned policy on the real robot.

### A. Biomimetic Eel-Like Robot Design

As shown in Fig. 1, the biomimetic robotic eel adopts a three-segment configuration, consisting of a front compliant body section, a central rigid control cabin, and a rear compliant body section connected in series, with a flexible caudal fin attached at the tail. Both the front and rear compliant body sections employ an Omnidirectional and Quadruple-Wire-Driven structure. Each compliant section is composed of multiple serially connected vertebral units, through which actuation cables pass and are driven by servo motors to generate continuous and coordinated bending motion along the body. Elastic restoring elements are installed between adjacent units, while the flexible caudal fin further enhances propulsion performance. A vision module is mounted at the head for environmental perception in autonomous tasks, whereas the central control cabin integrates the onboard computing unit, low-level controller, battery, and communication module, serving as the main computation and control unit of the robot. The main specifications of the robotic eel platform are listed in Table I.

The platform adopts an anguilliform body-wave propulsion pattern. Since the front body section is closer to the head, its deformation has a more direct influence on head deflection and heading variation. In contrast, the rear compliant section and the caudal structure mainly contribute to body-wave propagation and thrust generation. Therefore, the platform exhibits a functional division of front-end steering and rear-end propulsion in both structure and locomotion. This characteristic provides the basis for the subsequent AHTA-CPG control, making it more suitable to achieve relatively decoupled control of steering and propulsion while preserving the anguilliform body-wave locomotion pattern.

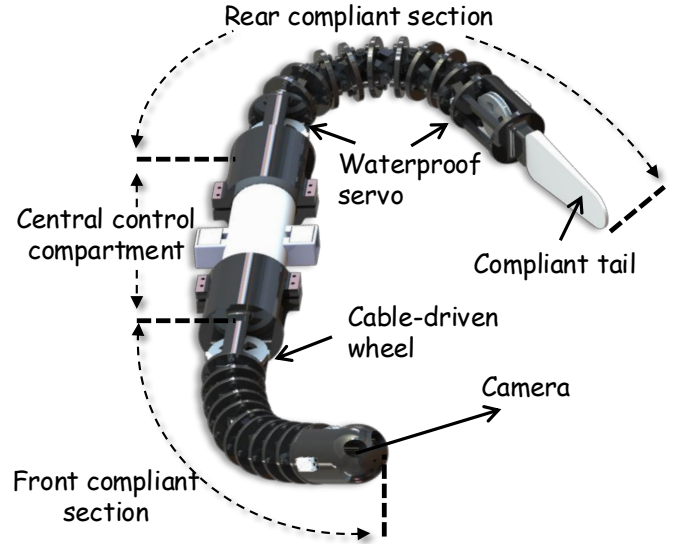


Fig. 1. Structural overview of the biomimetic eel-like robot. The platform consists of a front compliant section, a central control compartment, a rear compliant section, and a compliant tail. A waterproof servo and a cable-driven wheel are integrated for actuation, while a camera is mounted at the head for environmental perception.

TABLE I  
KEY SPECIFICATIONS OF THE BIOMIMETIC EEL-LIKE ROBOT PLATFORM

| Parameter              | Specification                             |
|------------------------|-------------------------------------------|
| Overall dimensions     | 1870 mm × 144 mm × 119 mm                 |
| Total mass             | 10.65 kg                                  |
| Servo specification    | Standard servo motor (6 V, 35 kg·cm)      |
| Onboard computing unit | Jetson Orin Nano Super Dev Kit            |
| Low-level controller   | STM32F103                                 |
| Battery                | Lithium-polymer battery (7.4 V, 2000 mAh) |

### B. MuJoCo-Based Simulation Environment

Considering that existing dedicated underwater simulators mainly target propeller-driven ROV/AUV platforms and provide limited support for the modeling and control of body-wave-propelled biomimetic eel-like robots, while autonomous obstacle avoidance also requires efficient simulation of contact, collision, and dynamic interaction, this work adopts MuJoCo together with simplified underwater dynamics to construct the simulation environment for subsequent policy training.

As shown in the left blue part of Fig. 2, the simulation model is constructed by simplifying the physical prototype. While preserving the key components of the robot, including the front compliant body section, the central rigid control cabin, and the rear compliant body section, the driving cables and joint structures are appropriately simplified to retain the main motion characteristics relevant to control while improving simulation efficiency. The robotic eel adopts an anguilliform body-wave propulsion pattern, whose lateral displacement can be approximately described as

$$y(x, t) = A(x) \sin \left( \frac{2\pi x}{\lambda} - 2\pi f t + \phi \right), \quad (1)$$

where  $A(x)$ ,  $\lambda$ ,  $f$ , and  $\phi$  denote the amplitude envelope, wave-length, oscillation frequency, and initial phase, respectively.

Based on the above motion representation, AHTA-CPG is introduced in simulation as a low-level control method matched to the locomotion mechanism of the platform. AHTA-CPG incorporates the head deflection angle into the frequency modulation of the tail rhythmic signal, with the key relationship given by

$$\dot{\phi} = \omega (1 + k \cdot \text{sgn}(\theta_{\text{head}}) \cdot |\theta_{\text{head}}|), \quad (2)$$

where  $\theta_{\text{head}}$  denotes the head deflection angle,  $\dot{\phi}$  denotes the phase rate of the tail rhythmic signal,  $\omega$  is the base oscillation frequency, and  $k$  is the modulation coefficient. This mechanism adjusts the tail oscillation frequency according to both the direction and magnitude of the head deflection angle, thereby establishing an explicit link between head steering and tail propulsion and making the control process more consistent with the biomimetic locomotion characteristics of the robotic eel.

Compared with directly outputting low-level servo commands, the introduction of AHTA-CPG allows the policy to avoid learning the complex body-wave generation process directly, and instead only adjust a small number of key control variables, thereby reducing the policy search space and improving training efficiency.

To describe the main hydrodynamic effects acting on the robotic eel in water, a simplified underwater dynamics model is further adopted for control-oriented training. Specifically, the fish body is divided into multiple hydrodynamic segments, and the tangential velocity  $u_i$  and normal velocity  $w_i$  of each segment relative to the surrounding water are computed. The hydrodynamic force acting on the  $i$ -th segment is expressed as

$$F_i = F_{t,i} + F_{n,i} + F_{a,i}, \quad (3)$$

where the three terms correspond to the tangential drag, normal drag, and normal added-mass force, respectively. These components are modeled as

$$F_{t,i} = -C_t u_i t_i, \quad (4)$$

$$F_{n,i} + F_{a,i} = -C_{n,i} |w_i| w_i n_i - m_{a,i} a_{n,i} n_i. \quad (5)$$

Here,  $t_i$  and  $n_i$  denote the tangential and normal unit vectors of the  $i$ -th segment,  $C_t$  and  $C_{n,i}$  are the tangential and normal drag coefficients,  $m_{a,i}$  is the added-mass coefficient, and  $a_{n,i}$  is the normal acceleration. In addition to the segment-wise hydrodynamic forces, a yaw damping term and an added rotational inertia term are introduced to approximate the hydrodynamic dissipation and inertial effects during turning. In this way, the model captures the dominant hydrodynamic influences on both propulsion and steering, while remaining sufficiently simple for efficient policy training in simulation.

### C. Two-Stage Learning for Autonomous Obstacle Avoidance

As shown in the training pipeline of Fig. 2, the proposed method adopts a two-stage learning framework for autonomous obstacle avoidance of the biomimetic eel-like

robot. In this task, the robot is required to avoid obstacles while reaching a predefined goal. The policy is first initialized through imitation learning and then further refined by reinforcement learning. By combining task-oriented policy learning with a control method consistent with the locomotion characteristics of the platform, the robotic eel can learn stable obstacle-avoidance behaviors in complex environments.

For this autonomous obstacle avoidance task, the observation is defined as

$$o_t = \{\mathbf{I}_t, \mathbf{m}_t, \mathbf{d}_t, \phi_t\}, \quad (6)$$

where  $I_t$  denotes the head-camera image,  $\mathbf{m}_t$  denotes the IMU measurements, and  $d_t$  and  $\phi_t$  denote the relative distance and bearing angle of the robot with respect to the goal, respectively. As defined in (6), the observation integrates visual perception, proprioceptive sensing, and goal-related guidance, thereby preserving the vision-based autonomous control setting while enabling the robot to move toward the goal while avoiding obstacles.

In the action design, the policy does not directly and independently output the control commands of both servos. Instead, it only outputs the key control variable related to steering, while the control signal of the other servo is generated according to the AHTA-CPG relationship. This design reduces the complexity of the action space and makes the control output more consistent with the biomimetic locomotion characteristics of the robotic eel.

To enable effective learning, the reward function for autonomous obstacle avoidance is defined as

$$r_t^{\text{avoid}} = \alpha_1 r_t^{\text{goal}} + \alpha_2 r_t^{\text{arrive}} - \alpha_3 r_t^{\text{collision}}, \quad (7)$$

where  $r_t^{\text{goal}}$  denotes the goal-approaching reward, encouraging the robot to move toward the goal;  $r_t^{\text{arrive}}$  denotes the goal-reaching reward, reinforcing task completion; and  $r_t^{\text{collision}}$  denotes the collision penalty, suppressing unsafe contact with obstacles. As defined in (7), the reward is designed to encourage the learned policy to balance task completion and safety.

In the two-stage training pipeline shown in Fig. 2, the first stage initializes the policy through imitation learning, using a small set of expert demonstrations collected from a few representative scenes sampled from the 100 training scenes used in reinforcement learning. In the second stage, reinforcement learning is employed to further fine-tune the initialized policy. Specifically, a fixed training set of 100 scenes is used in the reinforcement learning stage. In each training cycle, the policy interacts with each scene once to collect rollouts, and the aggregated trajectories are then used for one PPO update.

For policy optimization, Proximal Policy Optimization (PPO) is adopted. Let  $\theta$  denote the policy parameters. The clipped PPO objective is defined as

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (8)$$

where the probability ratio is given by

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \quad (9)$$

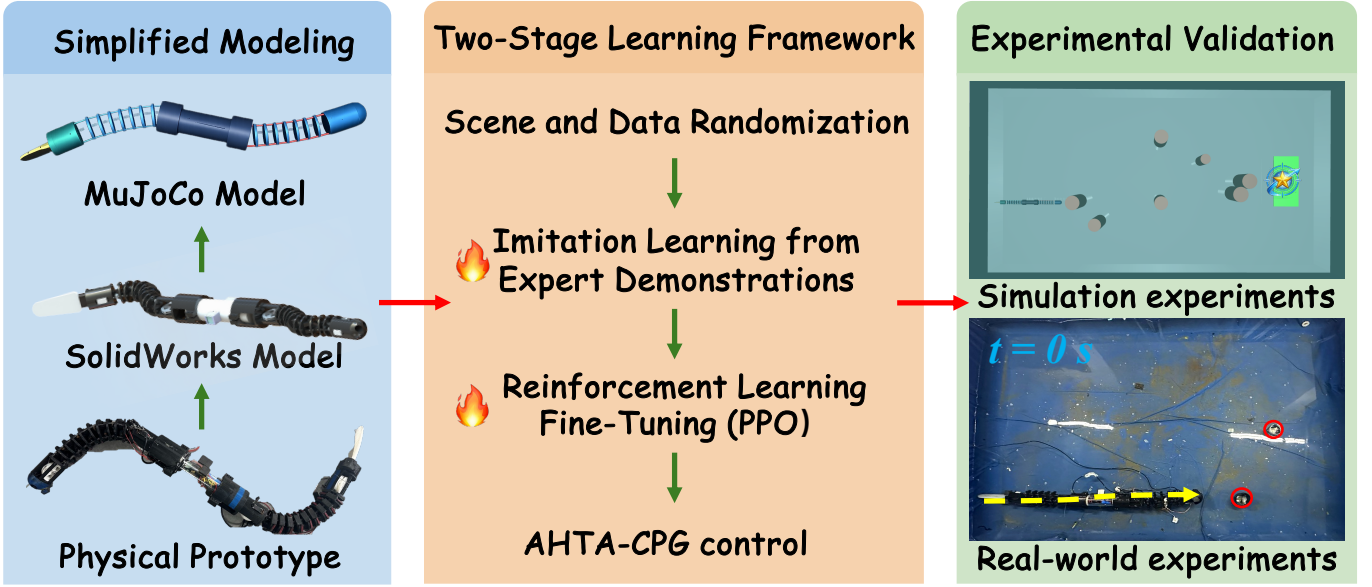


Fig. 2. Overview of the proposed two-stage learning framework for autonomous obstacle avoidance of the biomimetic eel-like robot. Starting from the physical prototype, a simplified simulation model is constructed for training. The policy is first initialized by imitation learning from expert demonstrations and then refined by PPO-based reinforcement learning, with the learned policy deployed through AHTA-CPG control. The framework is validated in both simulation and real-world experiments.

$\hat{A}_t$  is the estimated advantage, and  $\epsilon$  is the clipping coefficient. As defined in (8) and (9), PPO constrains the update magnitude between the new and old policies, thereby improving training stability and making it suitable for the continuous-control task considered in this work.

During training, the policy network outputs the steering-related control variable based on the observation in (6) and updates its parameters through continuous interaction with the environment to maximize the reward in (7). After training, the learned policy can achieve stable autonomous obstacle avoidance on the robotic eel platform while maintaining the ability to move toward the predefined goal.

#### D. Domain Randomization for Real-Robot Deployment

During policy training, obstacle avoidance scenes are randomly generated in simulation to increase scene diversity and improve policy generalization. Specifically, each scene consists of a water tank and randomly generated obstacles. Obstacles are placed at random positions within the tank, and the number of obstacles in each scene is randomly set between 6 and 8. In this way, a diverse set of training scenes is constructed for reinforcement learning.

In addition, to reduce the visual gap between simulated images and real underwater images, a visual domain randomization method based on real underwater imagery is further adopted. Specifically, real images collected from underwater environments are used as visual references, without requiring correspondence to specific task trajectories or strict one-to-one matching with simulated scenes. Before policy training, part of the simulated image samples are randomly varied according to the appearance characteristics observed in real underwater

images. This randomization mainly affects visual factors such as color distribution, brightness level, blur degradation, and background texture, thereby improving the robustness of the learned policy for deployment on the real biomimetic eel-like robot platform.

### III. EXPERIMENTS

#### A. Experimental Setup

For the two-stage learning framework, including both imitation learning and reinforcement learning fine-tuning, all model training was conducted on an NVIDIA A800 GPU.

The experiments were conducted in both simulation and real-world environments. In simulation, the obstacle-avoidance dataset was divided into a training set and a test set, each containing 100 scenes. During evaluation, 20 scenes were sampled from the corresponding dataset, and each scene was tested for 5 trials, resulting in 100 trials in total. In simulation, a trial was regarded as successful when the robot reached the goal and the number of consecutive collisions was less than 5. The performance was evaluated using the success rate (SR) and the degree of completion (DOC). DOC is defined as

$$\text{DOC} = 1 - \frac{\|\mathbf{p}_{\text{final}} - \mathbf{p}_{\text{goal}}\|}{\|\mathbf{p}_{\text{start}} - \mathbf{p}_{\text{goal}}\|} \quad (10)$$

where  $\mathbf{p}_{\text{start}}$ ,  $\mathbf{p}_{\text{final}}$ , and  $\mathbf{p}_{\text{goal}}$  denote the start position, final position, and goal position, respectively. A larger DOC indicates better task performance.

The real-world experiment was conducted in a water tank with 2–6 randomly placed obstacles. A total of 60 trials were performed to evaluate the practical deployment capability of the proposed method. In the real-world experiment, a trial was



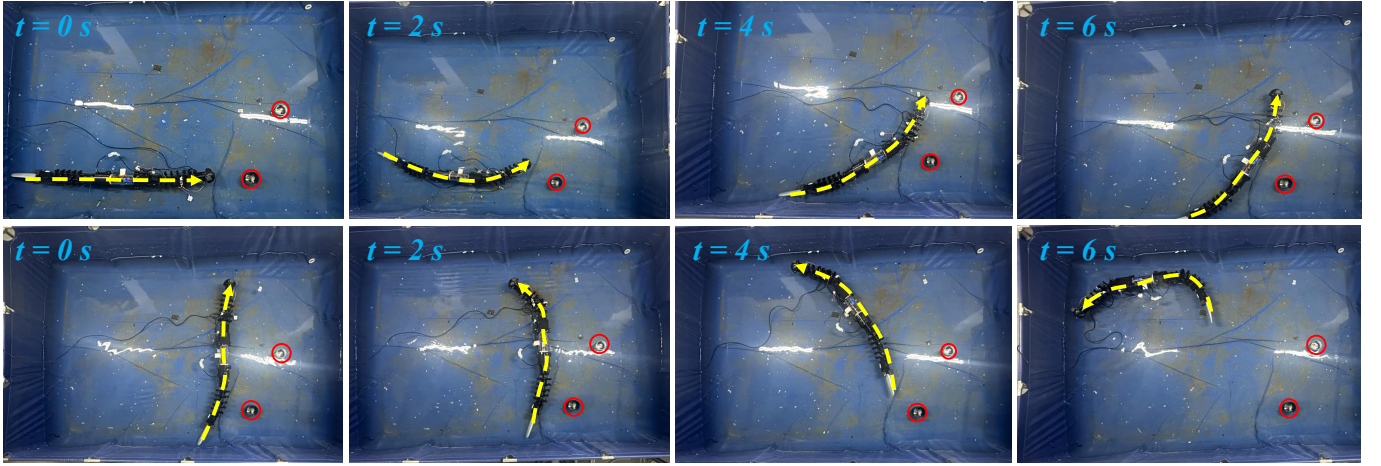


Fig. 3. Time-sequence snapshots of real-world autonomous obstacle avoidance by the biomimetic eel-like robot. The top row illustrates avoidance of static obstacles, while the bottom row shows wall-avoidance behavior near the tank walls. Red circles mark the obstacle locations.

regarded as successful only when the robot reached the goal without any collision.

### B. Effectiveness of Two-Stage Learning

To evaluate the proposed two-stage learning framework, we vary the IL data ratio used for policy initialization and then fine-tune the initialized policy with PPO. Six settings are considered, corresponding to 0%, 20%, 40%, 60%, 80%, and 100% of the available imitation-learning data, where 0% means PPO is trained from scratch. The results are summarized in Table II.

All policies are fine-tuned under the same reinforcement learning setting and evaluated on both the training set and the test set. For comparison, Table II also reports the performance of imitation learning alone under different data ratios, highlighting the necessity of the second-stage policy refinement.

As shown in Table II, the benefit of PPO fine-tuning gradually decreases as IL data ratio increases. The largest gain is observed at the 20% data ratio, indicating that a relatively small amount of expert data can already provide an effective policy initialization while preserving substantial room for further improvement through reinforcement learning. By contrast, a higher proportion of imitation-learning data leads to a stronger initial policy but smaller gains from subsequent reinforcement learning fine-tuning.

### C. Effectiveness of the Control Method

To verify the effectiveness of the proposed control framework, we compare three control methods: direct dual-servo control, traditional CPG control, and the proposed AHTA-CPG control. In the direct dual-servo control setting, the policy independently outputs the commands of both servos. In the traditional CPG control setting, the policy outputs the steering-related control variable, while the tail motion is generated by a CPG with a fixed frequency of 0.7 Hz. In the proposed AHTA-CPG control setting, the policy also outputs only the steering-related control variable, while the other servo

TABLE II  
RESULTS UNDER DIFFERENT DATA RATIOS AND TRAINING STRATEGIES

| IL data ratio | Method   | Train SR (%) | Train DOC     | Test SR (%) | Test DOC      |
|---------------|----------|--------------|---------------|-------------|---------------|
| 0%            | PPO only | 0            | 0.571         | 0           | 0.488         |
| 20%           | IL only  | 23           | 0.651         | 11          | 0.619         |
| 20%           | IL + PPO | 63(↑40)      | 0.815(↑0.164) | 55(↑44)     | 0.731(↑0.112) |
| 40%           | IL only  | 32           | 0.684         | 29          | 0.616         |
| 40%           | IL + PPO | 65(↑33)      | 0.827(↑0.143) | 50(↑21)     | 0.718(↑0.102) |
| 60%           | IL only  | 35           | 0.728         | 33          | 0.717         |
| 60%           | IL + PPO | 67(↑32)      | 0.824(↑0.096) | 47(↑14)     | 0.789(↑0.072) |
| 80%           | IL only  | 48           | 0.767         | 42          | 0.761         |
| 80%           | IL + PPO | 63(↑15)      | 0.803(↑0.036) | 51(↑9)      | 0.818(↑0.057) |
| 100%          | IL only  | 55           | 0.783         | 43          | 0.779         |
| 100%          | IL + PPO | 60(↑5)       | 0.795(↑0.012) | 49(↑6)      | 0.811(↑0.032) |

command is adaptively generated according to the AHTA-CPG relationship.

All three methods are trained under the same two-stage learning pipeline and evaluated using identical training and test protocols. This comparison is intended to show the influence of different control methods on the autonomous obstacle-avoidance performance of the biomimetic eel-like robot and to verify the effectiveness of the proposed control method.

TABLE III  
COMPARISON OF DIFFERENT CONTROL METHODS

| Control Method                       | Train SR (%) | Test SR (%) |
|--------------------------------------|--------------|-------------|
| Direct dual-servo control            | 37           | 29          |
| CPG (tail frequency fixed at 0.7 Hz) | 44           | 35          |
| AHTA-CPG(Ours)                       | 63           | 55          |

### D. Comparison of RL Fine-Tuning Algorithms

To compare different reinforcement learning algorithms in the proposed framework, we initialize the policy using the model trained with 20% imitation-learning data, and then fine-tune it using PPO, TRPO, and A2C, respectively. The resulting policies are evaluated on both the training set and the test set using the same protocol.

This experiment is used to identify which reinforcement learning algorithm is more suitable for fine-tuning the obstacle-avoidance policy of the biomimetic eel-like robot under the current setting.

TABLE IV  
COMPARISON OF DIFFERENT RL FINE-TUNING ALGORITHMS

| RL algorithm | Train SR (%) | Test SR (%) | Test DOC     |
|--------------|--------------|-------------|--------------|
| A2C          | 45           | 15          | 0.621        |
| TRPO         | 55           | 42          | 0.705        |
| PPO(Ours)    | <u>63</u>    | <u>55</u>   | <u>0.731</u> |

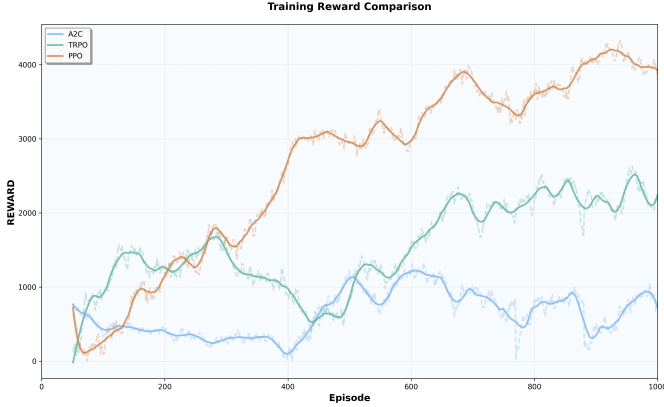


Fig. 4. Training reward comparison of different reinforcement learning fine-tuning algorithms initialized from the same imitation-learning model trained with 20% imitation-learning data. PPO achieves the highest reward and shows more stable training performance compared with TRPO and A2C.

#### E. Discussion on Reward Design

As defined in (7),  $r_t^{\text{arrive}}$  and  $r_t^{\text{collision}}$  are essential components required by the task itself, since the robot must reach the goal while avoiding collisions. The term  $r_t^{\text{goal}}$  is further introduced as a shaping reward to guide policy learning, and is defined as

$$r_t^{\text{goal}} = \frac{d_{t-1} - d_t}{d_0}, \quad (11)$$

where  $d_t$  denotes the Euclidean distance between the robot and the goal at time step  $t$ , and  $d_0$  denotes the initial distance between the start position and the goal. Without this term, the robotic eel was observed to circle around obstacles without effectively moving toward the goal, which is verified by simulation results. Therefore,  $r_t^{\text{goal}}$  helps provide directional guidance and improves training effectiveness.

#### F. Real-World Experiments

To evaluate the real-world deployment capability of the proposed method, we conduct experiments in a water tank with 2, 4, and 6 randomly placed obstacles. For each obstacle-number setting, 20 trials are performed. In each trial, the robotic eel is required to navigate from the initial position to the goal while avoiding collisions with obstacles and the tank wall. The real-world experiments focus on the final deployed policy obtained by the proposed two-stage learning

framework. A trial is counted as successful only when the robot reaches the goal without any collision. Compared with the simulated scenarios, the real-world experiments involve relatively smaller obstacles and a simpler setup, which makes the task less challenging and may partially account for the higher success rate observed in practice. To compensate for this difference, a stricter success criterion is adopted in the real-world experiments, where any collision leads to task failure. Nevertheless, the results still demonstrate the practical deployment capability of the proposed method on the physical robotic platform.

TABLE V  
REAL-WORLD EXPERIMENTAL RESULTS

| Number of obstacles | Trials | Real-world SR (%) |
|---------------------|--------|-------------------|
| 2                   | 20     | 90                |
| 4                   | 20     | 75                |
| 6                   | 20     | 70                |

#### IV. CONCLUSION

This paper proposes a two-stage learning framework for autonomous obstacle avoidance of a biomimetic eel-like robot. By combining imitation-learning-based policy initialization with reinforcement-learning-based policy fine-tuning, the proposed framework improves policy learning effectiveness for this strongly coupled biomimetic platform. In addition, by incorporating AHTA-CPG as the low-level control method, the learned policy becomes better aligned with the locomotion characteristics of the robot. Both simulation and real-world experiments validate the effectiveness of the proposed framework and demonstrate its practical deployment capability in obstacle-avoidance tasks.

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